SUMIT BANSIYA

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PROFESSIONAL SUMMARY

AI Researcher and M.Tech Scholar specializing in Computer Vision and Generative AI. Proven track record in developing high-fidelity medical imaging models (GANs) and optimizing deep learning pipelines for real-world deployment. Skilled in building RAG-based LLM applications and engineering scalable inference systems using Python and modern MLOps tools. Passionate about solving complex healthcare and industrial problems through applied AI.

TECHNICAL SKILLS

- **Languages:** Python, Object-Oriented Programming
- **Machine Learning:** Feature Engineering, Statistical Analysis, Hyperparameter Tuning (Optuna), Model Evaluation
- **Deep Learning:** CNNs (ResNet, EfficientNet), GANs (Pix2Pix), Transformers, Transfer Learning
- **Generative AI:** LLMs, RAG Pipelines, Prompt Engineering, Vector Embeddings, Stable Diffusion
- **Frameworks & Libraries:** PyTorch, TensorFlow, OpenCV, Scikit-learn, Hugging Face
- **Tools & Deployment:** Git, VS Code, Google Colab, FastAPI, Streamlit, Jupyter

PROFESSIONAL EXPERIENCE

AI Prompt Engineer (Freelance) — Outlier AI Oct 2024 – Present

- Optimizing prompt strategies for Large Language Models (LLMs) to enhance reasoning capabilities in complex analytical and coding tasks.
- Conducting rigorous RLHF (Reinforcement Learning from Human Feedback) evaluations to improve factual accuracy and safety protocols.
- Refining mathematical reasoning chains, resulting in higher fidelity outputs for multi-step problem-solving queries.

Machine Learning Intern — Persistent Systems Jun 2023 – Aug 2023

- Engineered automated data preprocessing pipelines using Python (Pandas/NumPy), significantly reducing data cleaning time for large-scale datasets.
- Collaborated on the integration of ML-based components into production environments, ensuring code modularity and runtime efficiency.

KEY PROJECTS

Generative AI for Medical Imaging (Research Project)

- Designed a Pix2Pix GAN architecture to synthesize Dual-Energy Subtracted (DES) phantoms from low-energy mammograms, facilitating radiation-free screening research.
- Achieved high structural fidelity, validated via quantitative metrics (PSNR, SSIM, LPIPS) and ROI-based statistical analysis.
- Currently optimizing the pipeline for high-resolution lesion reconstruction and artifact removal.

Car Damage Assessment System (Deployment & Optimization)

- Developed an end-to-end damage classification system using a custom dataset of 2,300 vehicle images.
- Optimized model performance from 57% to 80% accuracy by transitioning from EfficientNet-B0 to a ResNet backbone.

- Implemented automated hyperparameter tuning using Optuna (optimizing Learning Rate and Dropout).
- Deployed the model as a real-time inference service using FastAPI (backend) and Streamlit (frontend).

Explainable Multi-View Breast Cancer Diagnosis

- Built a multi-view deep learning framework utilizing 8 CESM views for granular Normal/Benign/Malignant classification.
- Integrated Grad-CAM for visual explainability, ensuring model decisions align with radiological features.

Advanced Multi-Stage RAG Pipeline for Legal Document QA

- Architected a production-grade RAG system using hybrid retrieval (BM25 + FAISS) to maximize lexical and semantic coverage for legal documents.
- Implemented LLM-based query rewriting to generate structured sub-queries, improving retrieval recall by approximately 22%.
- Integrated cross-encoder reranking (MiniLM) to filter irrelevant contexts and enhance answer precision.
- Applied context compression and adaptive chunking to reduce token usage and improve response coherence.
- Deployed as a FastAPI service with Redis caching, reducing latency from 4.2s to 0.9s for repeated queries.

Agentic AI Research & Report Automation System

- Designed a multi-agent system using CrewAI to automate research and competitive intelligence workflows.
- Developed specialized agents for research, analysis, fact-checking, and report generation with structured orchestration.
- Integrated real-time search (Serper API) and LangChain-based reasoning for multi-hop information synthesis.
- Implemented human-in-the-loop checkpoints to ensure output reliability and domain correctness.
- Generated comprehensive reports (12+ pages) within minutes, reducing manual effort by over 70%.

EDUCATION

M.Tech in Artificial Intelligence — NIT Jalandhar 2025 – Present
CGPA: 8.43 — Focus: Medical Imaging, Deep Learning Research

B.Tech in IT (Artificial Intelligence) — MITS Gwalior 2020 – 2024
CGPA: 8.7

CERTIFICATIONS

- Generative AI with AWS — Udacity
- Python for Data Science and Machine Learning — Udemy
- NPTEL: Advanced Graph Theory (Elite + Silver), Google Cloud Computing (Elite), Natural Hazards (Elite), NLP, Social Network Analysis